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No:

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**2025-2026 Öğretim Yılı Bahar Dönemi**

**MÜHENDİSLİK FAKÜLTESİ GENEL MATEMATİK II ARASINAVI**

**(Süre 80 dk.)**

**1.** $\lim_{x\to 0}\dfrac{\int_{x^2}^{0}\sin t^2\,dt}{2x^6}=?$ **(20 p.)** $\frac{0}{0}$

$$\lim_{x\to 0}\frac{-2x\cdot\sin x^4}{12x^5}=\lim_{x\to 0}\frac{-2\cdot\sin x^4}{12\cdot x^4}=-\frac{1}{6}$$ /10

**2.** $\int_{-\infty}^{2}\dfrac{x^2}{1+x^3}dx$ integralinin yakınsaklık durumunu inceleyiniz. Yakınsak ise değerini bulunuz. **(20 p.)**

$$I=\underbrace{\int_{-\infty}^{-2}\frac{x^2}{1+x^3}dx}_{I_1}+\underbrace{\int_{-2}^{-1}\frac{x^2}{1+x^3}dx}_{I_2}+\underbrace{\int_{-1}^{2}\frac{x^2}{1+x^3}dx}_{I_3}$$ 5/

ve $\frac{1}{3}\int\frac{3x^2}{1+x^3}dx=\frac{1}{3}\ln|x^3+1|$ 5/

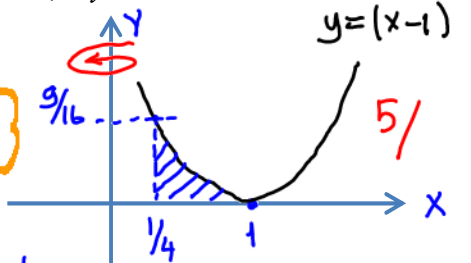
$$I_1=\lim_{a\to-\infty}\int_a^{-2}\frac{x^2}{1+x^3}dx=\lim_{a\to-\infty}\frac{1}{3}\ln|x^3+1|\Big|_a^{-2}=\lim_{a\to-\infty}\frac{1}{3}\{\ln 7-\ln|a^3+1|\}$$ 5/

$$=\frac{1}{3}\ln 7-\infty=-\infty$$

$I_1$ integrali iraksak. Diğerlerine bakmaya gerek yok.

o halde $I$ integrali iraksaktır. 5/

**3.** $y=(x-1)^2$ parabolü, $x=\dfrac{1}{4}$ doğrusu ve $Ox$ ekseni ile sınırlı bölgeyi çizerek, $Oy$ ekseni etrafında döndürülmesi ile elde edilen dönel cismin hacmini bulunuz. **(20 p.)**




5/

Disk Metodu ile;

$(x-1)=\mp\sqrt{y}$

$x=1\mp\sqrt{y}$

$$V_y=\pi\int_0^{9/16}[(1-\sqrt{y})^2-(\tfrac{1}{4})^2]dy$$ 5/

$$=\pi\int_0^{9/16}(1-2\sqrt{y}+y-\tfrac{1}{16})dy=\pi\int_0^{9/16}(\tfrac{15}{16}-2\cdot y^{1/2}+y)dy$$ 5/

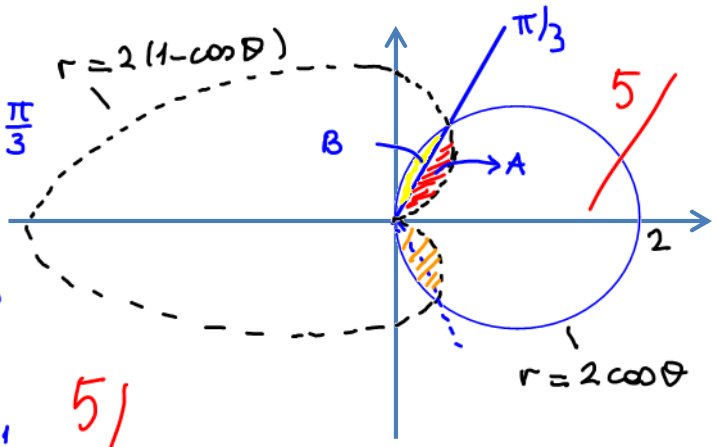
$$=\pi\left(\frac{15}{16}y-\frac{4}{3}y^{3/2}+\frac{y^2}{2}\right)\Big|_0^{9/16}=\pi\left(\frac{15\cdot 9}{16\cdot 16}-\frac{9}{16\cdot 2}+\frac{81}{16\cdot 16\cdot 2}\right)=\pi\frac{15\cdot 9\cdot 2-9\cdot 16\cdot 2+81}{16\cdot 16\cdot 2}=\frac{63\pi}{512}$$ 5/

Kabuk Metodu ile;

$$V_y=2\pi\int_{1/4}^{1}x\cdot(x^2-2x+1)dx=2\pi\left(\frac{x^4}{4}-\frac{2x^3}{3}+\frac{x^2}{2}\right)\Big|_{1/4}^{1}$$ 5/ 5/

$$=2\pi\left[\left(\tfrac{1}{4}-\tfrac{2}{3}+\tfrac{1}{2}\right)-\left((\tfrac{1}{4})^5-\frac{2}{3\cdot 4^3}+\frac{1}{16\cdot 2}\right)\right]=\frac{63\pi}{512}$$ 5/

**4.** $r = 2\cos\theta$ çemberinin ve $r = 2(1-\cos\theta)$ kardioidinin her ikisinin de içinde kalan bölgeyi çizerek alanını hesaplayınız. **(20 p.)**




$2(1-\cos\theta) = 2\cos\theta$

$\cos\theta = 1/2 \Rightarrow \theta = \mp\frac{\pi}{3}$

$$A = \frac{1}{2}\int_0^{\pi/3} 4(1-2\cos\theta+\underbrace{\cos^2\theta}_{\frac{1}{2}(1+\cos 2\theta)})\,d\theta$$

$$= 2\left(\frac{3}{2}\theta - 2\sin\theta + \frac{1}{4}\sin 2\theta\right)\Big|_0^{\pi/3}$$

$$= 2\left(\frac{\pi}{2} - \sqrt{3} + \frac{1}{4} + \frac{\sqrt{3}}{8}\right) = \pi - \frac{7\sqrt{3}}{4} + \frac{1}{4}$$ 5/

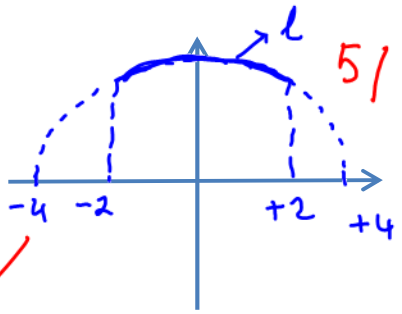
$$B = \frac{1}{2}\int_{\pi/3}^{\pi/2} 4\cdot\cos^2\theta = 2\int_{\pi/3}^{\pi/2} \frac{1}{2}(1+\cos 2\theta) = \theta + \frac{1}{2}\sin 2\theta\Big|_{\pi/3}^{\pi/2}$$

$$= \left(\frac{\pi}{2} - \frac{\pi}{3} - \frac{1}{2}\cdot\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6} - \frac{\sqrt{3}}{4}$$ 5/

$$\text{Alan} = 2(A+B) = 2\left\{\pi - \frac{7\sqrt{3}}{4} + \frac{1}{4} + \frac{\pi}{6} - \frac{\sqrt{3}}{4}\right\} = 2\cdot\left\{\frac{7\pi}{6} + \frac{1}{4} - 2\sqrt{3}\right\}$$ 5/

**5.** $y = \sqrt{16-x^2}$ eğrisinin $[-2,2]$ aralığındaki parçasını çizerek, Ox-ekseni etrafında döndürülmesiyle oluşan cismin yüzey alanını bulunuz. **(20 p.)**

$$S_x = 2\pi\int_{-2}^{+2} f(x)\cdot\sqrt{1+(f'(x))^2}\,dx$$




5/

$$1+(f'(x))^2 = 1+\left(\frac{x}{\sqrt{16-x^2}}\right)^2 = 1+\frac{x^2}{16-x^2} = \frac{16}{16-x^2}$$ 5/

$$S_x = 2\pi\int_{-2}^{+2} \sqrt{16-x^2}\cdot\frac{4}{\sqrt{16-x^2}}\,dx$$ 5/

$$= 2\pi\cdot 4x\Big|_{-2}^{+2} = 8\pi\cdot 4$$

$$= 32\pi \text{ br}^2$$ 5/